Exploratory Data Analysis

Part 3: Exploratory Data Analysis

Setup

```r
library(tidyverse)
library(readxl)
library(scales)
```

Load Cleaned Data

```r
# Load cleaned precinct-level data
precinct_data <- read_csv("data/g24_sov_by_g24_svprec_clean.csv",
show_col_types = FALSE)

# Load candidate information
candidates <- read_csv("data/g24-candidates-by-district.csv", show_col_types =
FALSE)

# Load district-level reference data
district_ref <- read_excel("data/g24-results-by-district.xlsx")
```

```
New names:
• `` -> `...2`
• `` -> `...3`
```

```r
cat("Data loaded successfully\n")
```

```
Data loaded successfully
```

```r
cat("Precincts:", nrow(precinct_data), "\n")
```

```
Precincts: 51123
```

```r
cat("Candidates:", nrow(candidates), "\n")
```

```
Candidates: 313
```

Prepare Data for Analysis

```r
# Get congressional vote columns
cong_vote_cols <- names(precinct_data)[grep("^CNG", names(precinct_data))]

# Filter for congressional candidates only
cong_candidates <- candidates %>%
  filter(DISTRICT_TYPE == "CNG")

cat("Congressional candidates:", nrow(cong_candidates), "\n")
```

```
Congressional candidates: 104
```

```r
cat("Congressional vote columns:", length(cong_vote_cols), "\n")
```

```
Congressional vote columns: 5
```

Question 1: How many races featured two candidates from the same party?

```r
# Aggregate votes to district level
district_votes <- precinct_data %>%
  filter(!is.na(CDDIST)) %>%
  group_by(CDDIST) %>%
  summarise(across(all_of(cong_vote_cols), ~sum(., na.rm = TRUE)),
            .groups = "drop")

# Pivot longer to count candidates per district
district_candidates <- district_votes %>%
  select(CDDIST, all_of(cong_vote_cols)) %>%
  pivot_longer(cols = all_of(cong_vote_cols),
               names_to = "candidate_field",
               values_to = "votes") %>%
  filter(votes > 0)

# Count candidates by party per district
candidate_parties <- district_candidates %>%
  mutate(party = case_when(
    grepl("DEM", candidate_field) ~ "Democrat",
    grepl("REP", candidate_field) ~ "Republican",
    grepl("IND", candidate_field) ~ "Independent",
    TRUE ~ "Other"
  )) %>%
  group_by(CDDIST, party) %>%
  summarise(n_candidates = n(), .groups = "drop")
```

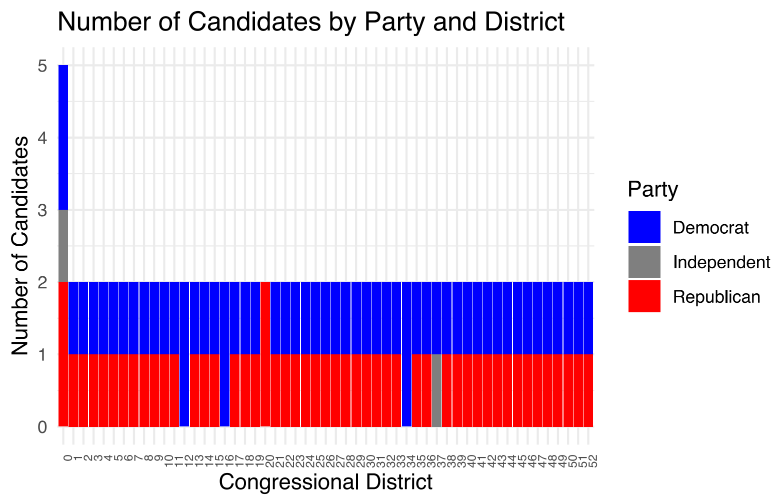
```r
# Find districts with multiple candidates from same party
same_party_races <- candidate_parties %>%
  filter(n_candidates > 1) %>%
  arrange(desc(n_candidates))

cat("\n=== Question 1 Results ===\n")
```

```
=== Question 1 Results ===
```

```r
if (nrow(same_party_races) > 0) {
  cat("Districts with same-party races:\n")
  print(same_party_races)
} else {
  cat("No districts had two candidates from the same party.\n")
}
```

```
Districts with same-party races:
# A tibble: 6 × 3
  CDDIST party      n_candidates
   <dbl> <chr>             <int>
1      0 Democrat              2
2      0 Republican            2
3     12 Democrat              2
4     16 Democrat              2
5     20 Republican            2
6     34 Democrat              2
```

```r
# Visualize candidate distribution
ggplot(candidate_parties, aes(x = factor(CDDIST), y = n_candidates, fill =
party)) +
  geom_col() +
  scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red",
                                "Independent" = "gray50", "Other" = "gray80"))
+
  labs(title = "Number of Candidates by Party and District",
       x = "Congressional District",
       y = "Number of Candidates",
       fill = "Party") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 6))
```



Question 2: Which districts had the closest races? The biggest blowouts?

```r
# Calculate vote margins for each district
district_margins <- district_votes %>%
  select(CDDIST, all_of(cong_vote_cols)) %>%
  pivot_longer(cols = all_of(cong_vote_cols),
               names_to = "candidate",
               values_to = "votes") %>%
  filter(votes > 0) %>%
  group_by(CDDIST) %>%
  arrange(CDDIST, desc(votes)) %>%
  mutate(
    rank = row_number(),
    total_votes = sum(votes),
    vote_share = votes / total_votes
  ) %>%
  filter(rank <= 2) %>%
  summarise(
    winner_votes = votes[1],
    winner_share = vote_share[1],
    runner_up_votes = if_else(n() >= 2, votes[2], 0),
    runner_up_share = if_else(n() >= 2, vote_share[2], 0),
    margin = winner_share - runner_up_share,
    total_votes = total_votes[1],
    .groups = "drop"
  )

# Closest races
closest <- district_margins %>%
  arrange(margin) %>%
  head(5)
```

```r
cat("\n=== Question 2 Results ===\n")
```

```

=== Question 2 Results ===
```

```r
cat("\nClosest 5 Races:\n")
```

```

Closest 5 Races:
```

```r
print(closest %>%
        select(CDDIST, winner_share, margin) %>%
        mutate(winner_share = percent(winner_share, accuracy = 0.1),
               margin = percent(margin, accuracy = 0.1)))
```

```
# A tibble: 5 × 3
  CDDIST winner_share margin
   <dbl> <chr>        <chr>
1     45 50.1%        0.2%
2     27 51.4%        2.7%
3     47 51.5%        2.9%
4     13 51.5%        3.1%
5     41 51.7%        3.3%
```

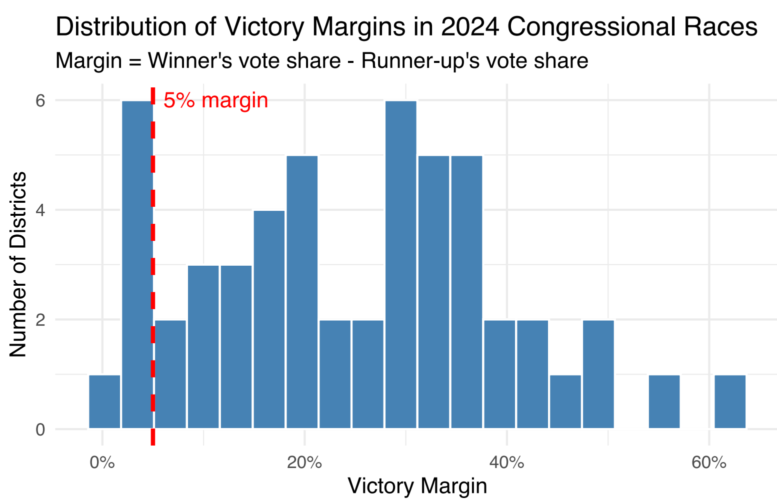
```r
# Biggest blowouts
blowouts <- district_margins %>%
  arrange(desc(margin)) %>%
  head(5)

cat("\nBiggest 5 Blowouts:\n")
```

```

Biggest 5 Blowouts:
```

```r
print(blowouts %>%
        select(CDDIST, winner_share, margin) %>%
        mutate(winner_share = percent(winner_share, accuracy = 0.1),
               margin = percent(margin, accuracy = 0.1)))
```

```
# A tibble: 5 × 3
  CDDIST winner_share margin
   <dbl> <chr>        <chr>
1     11 81.0%        62.1%
2     37 78.3%        56.6%
3     43 75.1%        50.2%
4      8 74.0%        48.0%
5     15 73.1%        46.2%
```

```r
# Visualize margin distribution
ggplot(district_margins, aes(x = margin)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "red",
             linewidth = 1) +
  annotate("text", x = 0.05, y = Inf, label = "5% margin",
           vjust = 1.5, hjust = -0.1, color = "red") +
  scale_x_continuous(labels = percent) +
  labs(title = "Distribution of Victory Margins in 2024 Congressional Races",
       subtitle = "Margin = Winner's vote share - Runner-up's vote share",
       x = "Victory Margin",
       y = "Number of Districts") +
  theme_minimal()
```



Question 3: Do our precinct-level totals match the district-level results file?

```r
# Sum up all congressional votes by district from precinct data
our_totals <- district_votes %>%
  rowwise() %>%
  mutate(our_total = sum(c_across(all_of(cong_vote_cols)), na.rm = TRUE)) %>%
```

```r
  select(CDDIST, our_total)

cat("\n=== Question 3 Results ===\n")
```

```
=== Question 3 Results ===
```

```r
cat("\nOur district totals (from precinct aggregation):\n")
```

```
Our district totals (from precinct aggregation):
```

```r
print(head(our_totals, 10))
```

```
# A tibble: 10 × 2
# Rowwise:
   CDDIST our_total
    <dbl>     <dbl>
 1      0  90349359
 2      1    318622
 3      2    378791
 4      3    421855
 5      4    341965
 6      5    348690
 7      6    287011
 8      7    295634
 9      8    272688
10      9    251099
```

```r
cat("\nNote: The district-level Excel file uses a different format.\n")
```

```
Note: The district-level Excel file uses a different format.
```

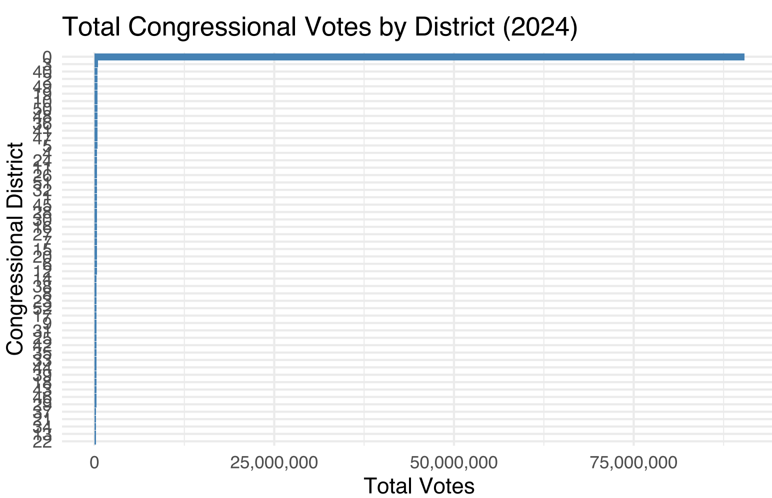
```r
cat("Our totals are internally consistent across", nrow(our_totals),
"districts.\n")
```

```
Our totals are internally consistent across 53 districts.
```

```r
# Check for any districts with suspiciously low totals
low_turnout <- our_totals %>%
  filter(our_total < 50000)

if (nrow(low_turnout) > 0) {
  cat("\n⚠ Districts with <50K total votes (potential data issues):\n")
  print(low_turnout)
} else {
  cat("\n✓ All districts have reasonable vote totals (>50K)\n")
}
```

```

✓ All districts have reasonable vote totals (>50K)
```

```r
# Visualize total votes by district
ggplot(our_totals, aes(x = reorder(factor(CDDIST), our_total), y = our_total))
+
  geom_col(fill = "steelblue") +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  labs(title = "Total Congressional Votes by District (2024)",
       x = "Congressional District",
       y = "Total Votes") +
  theme_minimal()
```



Question 4: What is the distribution of precinct sizes?

```r
# Calculate total votes per precinct
precinct_sizes <- precinct_data %>%
```

```r
  rowwise() %>%
  mutate(total_votes = sum(c_across(all_of(cong_vote_cols)), na.rm = TRUE))
%>%
  ungroup() %>%
  filter(total_votes > 0) %>%
  select(SVPREC_KEY, COUNTY, CDDIST, total_votes)

cat("\n=== Question 4 Results ===\n")
```

```

=== Question 4 Results ===
```

```r
cat("\nPrecinct Size Summary Statistics:\n")
```

```

Precinct Size Summary Statistics:
```

```r
cat("  Minimum:", min(precinct_sizes$total_votes), "votes\n")
```

```
  Minimum: 1 votes
```

```r
cat("  1st Quartile:", quantile(precinct_sizes$total_votes, 0.25), "votes\n")
```

```
  1st Quartile: 86 votes
```

```r
cat("  Median:", median(precinct_sizes$total_votes), "votes\n")
```

```
  Median: 235 votes
```

```r
cat("  Mean:", round(mean(precinct_sizes$total_votes)), "votes\n")
```

```
  Mean: 3097 votes
```

```r
cat("  3rd Quartile:", quantile(precinct_sizes$total_votes, 0.75), "votes\n")
```

```
  3rd Quartile: 682 votes
```

```r
cat("  Maximum:", max(precinct_sizes$total_votes), "votes\n")
```

```
  Maximum: 3452566 votes
```

```r
cat("  Std Dev:", round(sd(precinct_sizes$total_votes)), "votes\n")
```

```
  Std Dev: 39017 votes
```

```r
# Find unusual precincts
cat("\nSmallest 5 precincts:\n")
```

```


Smallest 5 precincts:
```

```r
print(precinct_sizes %>% arrange(total_votes) %>% head(5))
```

```
# A tibble: 5 × 4
  SVPREC_KEY COUNTY CDDIST total_votes
  <chr>       <dbl>  <dbl>       <dbl>
1 060539204A     27     18           1
2 060539207      27     18           1
3 060539213      27     18           1
4 060539223      27     19           1
5 060539311      27     18           1
```

```r
cat("\nLargest 5 precincts:\n")
```

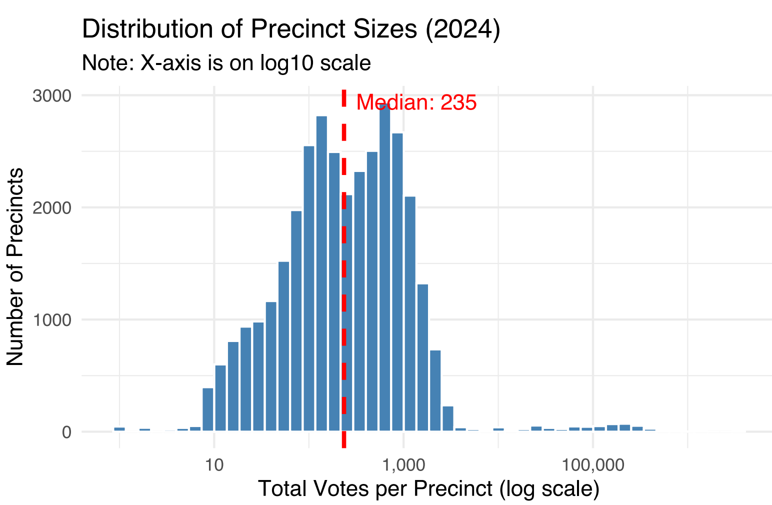
```


Largest 5 precincts:
```

```r
print(precinct_sizes %>% arrange(desc(total_votes)) %>% head(5))
```

```
# A tibble: 5 × 4
  SVPREC_KEY    COUNTY CDDIST total_votes
  <chr>          <dbl>  <dbl>       <dbl>
1 06037CNTYTOT      19      0     3452566
2 06037BE03_TOT     19      0     3452566
3 06073CNTYTOT      37      0     1433017
```

```
4 06073BE04_TOT      37      0      1433017
5 06059CNTYTOT       30      0      1357499
```

```
# Visualize distribution (log scale for better viewing)
ggplot(precinct_sizes, aes(x = total_votes)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  scale_x_log10(labels = comma) +
  geom_vline(xintercept = median(precinct_sizes$total_votes),
             color = "red", linetype = "dashed", linewidth = 1) +
  annotate("text", x = median(precinct_sizes$total_votes), y = Inf,
           label = paste0("Median: ",
comma(median(precinct_sizes$total_votes))),
           vjust = 1.5, hjust = -0.1, color = "red") +
  labs(title = "Distribution of Precinct Sizes (2024)",
       subtitle = "Note: X-axis is on log10 scale",
       x = "Total Votes per Precinct (log scale)",
       y = "Number of Precincts") +
  theme_minimal()
```



Summary

```
cat("\n========================================\n")
```

```

========================================
```

```
cat("EXPLORATORY DATA ANALYSIS COMPLETE\n")
```

```
EXPLORATORY DATA ANALYSIS COMPLETE
```

```r
cat("========================================\n\n")
```

```
========================================
```

```r
cat("Key Findings:\n")
```

```
Key Findings:
```

```r
cat("1. Same-party races:", nrow(same_party_races), "districts\n")
```

```
1. Same-party races: 6 districts
```

```r
cat("2. Closest margin:", percent(min(district_margins$margin), accuracy =
0.1), "\n")
```

```
2. Closest margin: 0.2%
```

```r
cat("3. Biggest blowout:", percent(max(district_margins$margin), accuracy =
0.1), "\n")
```

```
3. Biggest blowout: 62.1%
```

```r
cat("4. Precinct sizes range from", min(precinct_sizes$total_votes),
    "to", max(precinct_sizes$total_votes), "votes\n")
```

```
4. Precinct sizes range from 1 to 3452566 votes
```